Attendance QR code:

Texts that have been used to generate this lecture content

- Molecular Cell Biology Lodish et al
- Molecular Biology of the Cell Alberts et al
- Principles of Biochemistry Horton et al
- Molecular Biology of the Gene Watson et al

BIOC0001

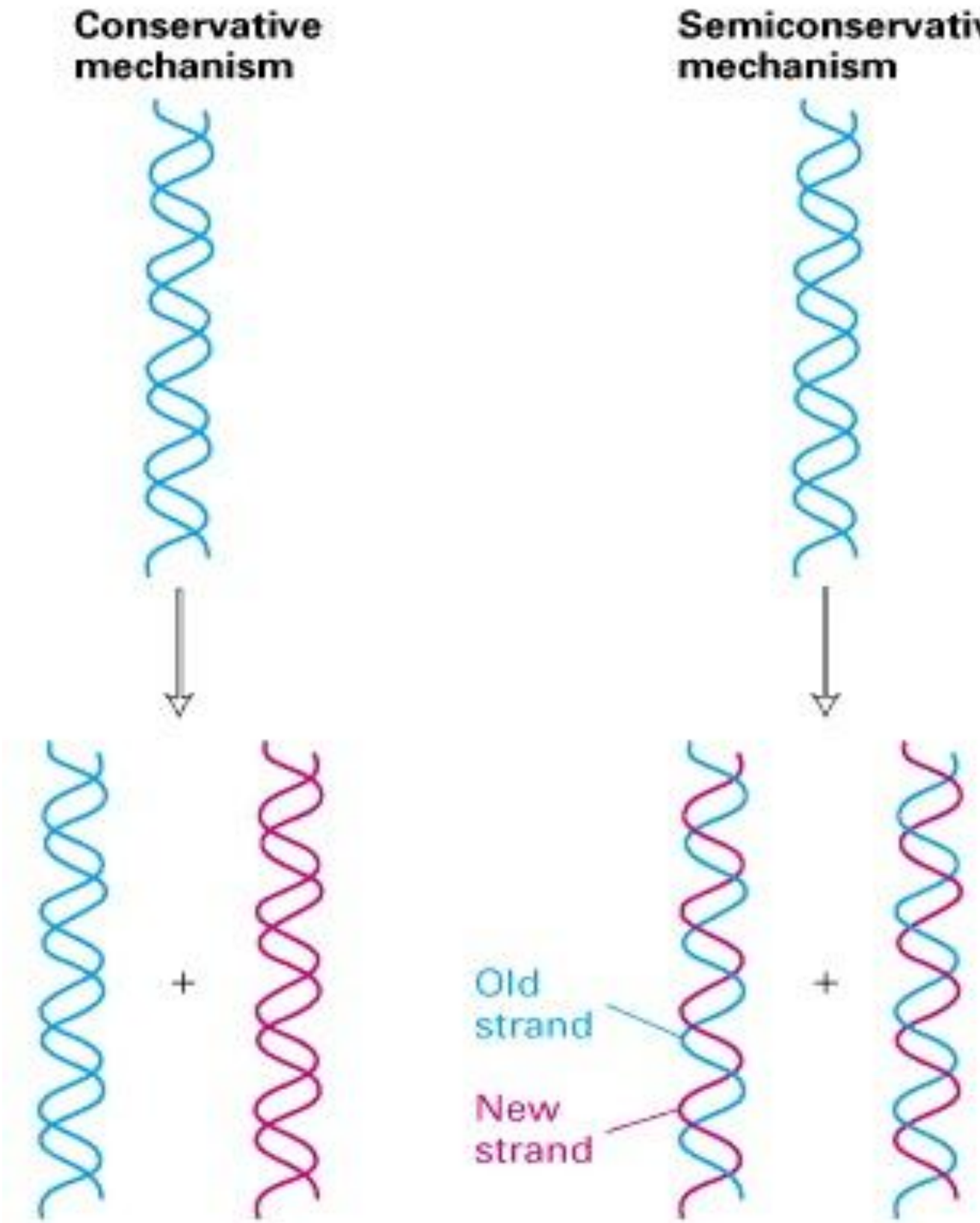
Lecture 3

DNA replication

- **Direction of replication**
- **Signals and proteins for starting replication**
- **DNA replication general mechanism**
- **Terminating DNA replication**
- **Correcting errors during DNA replication**
- **Application of DNA separation, re-annealing and replication in polymerase chain reaction (PCR)**

Dr. Suzanne Ruddy

Replication is semi-conservative



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Replication is bidirectional

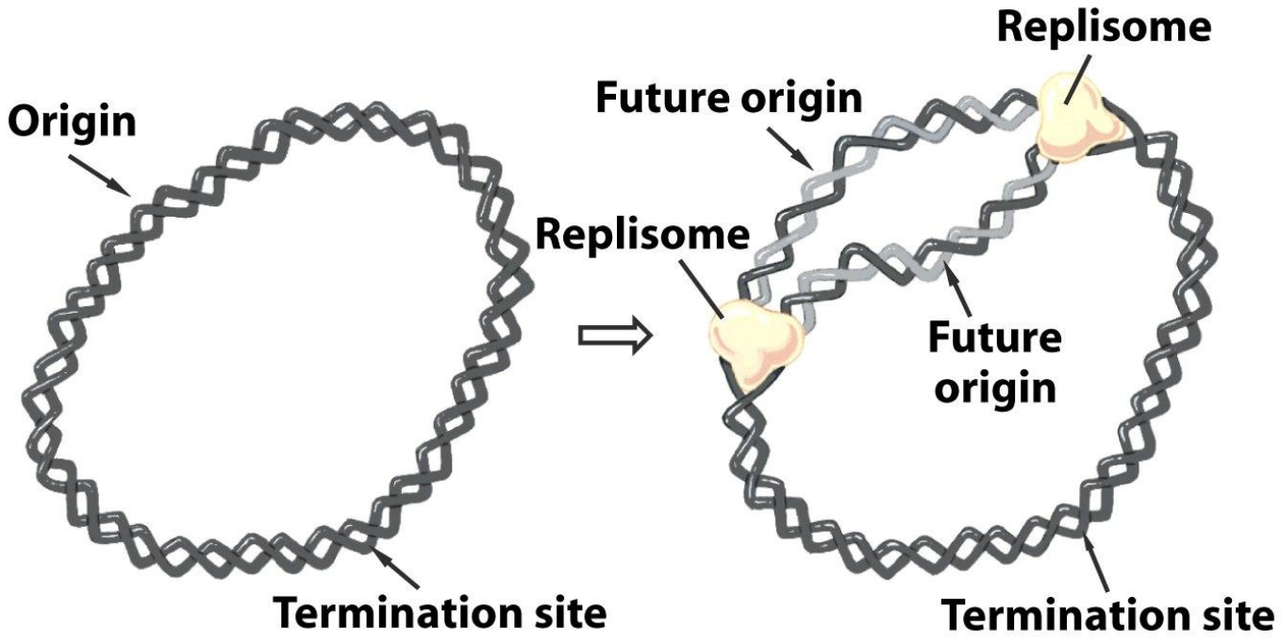


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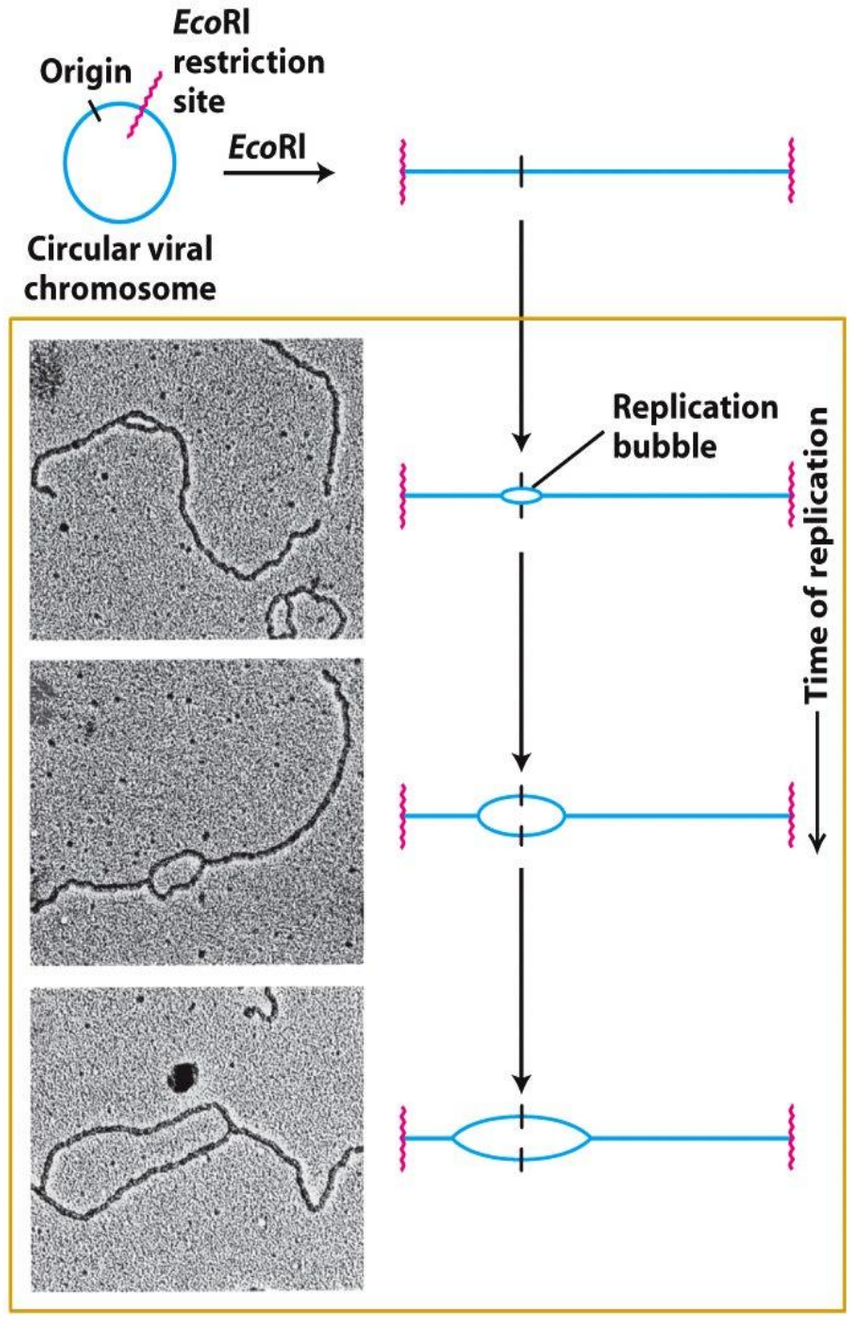


Figure 4-32
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Bidirectional

2 replication forks/origin

Where to begin replication? Prokaryotes

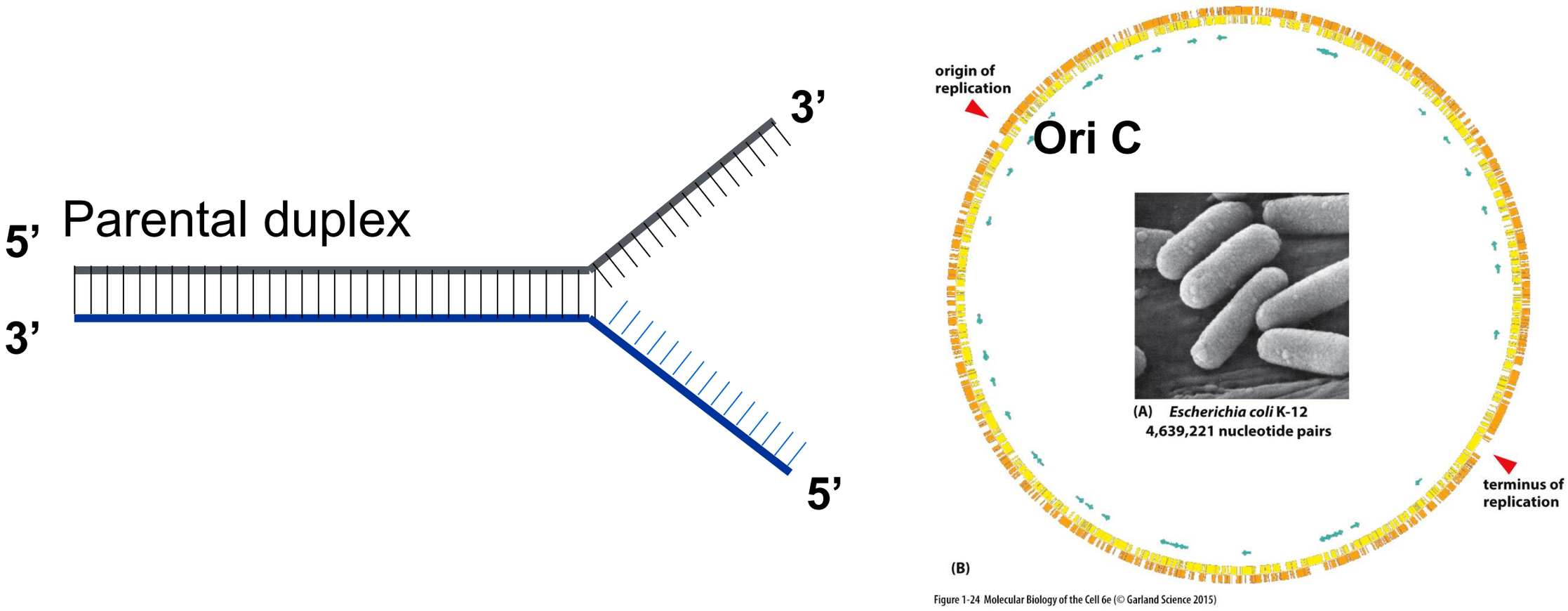
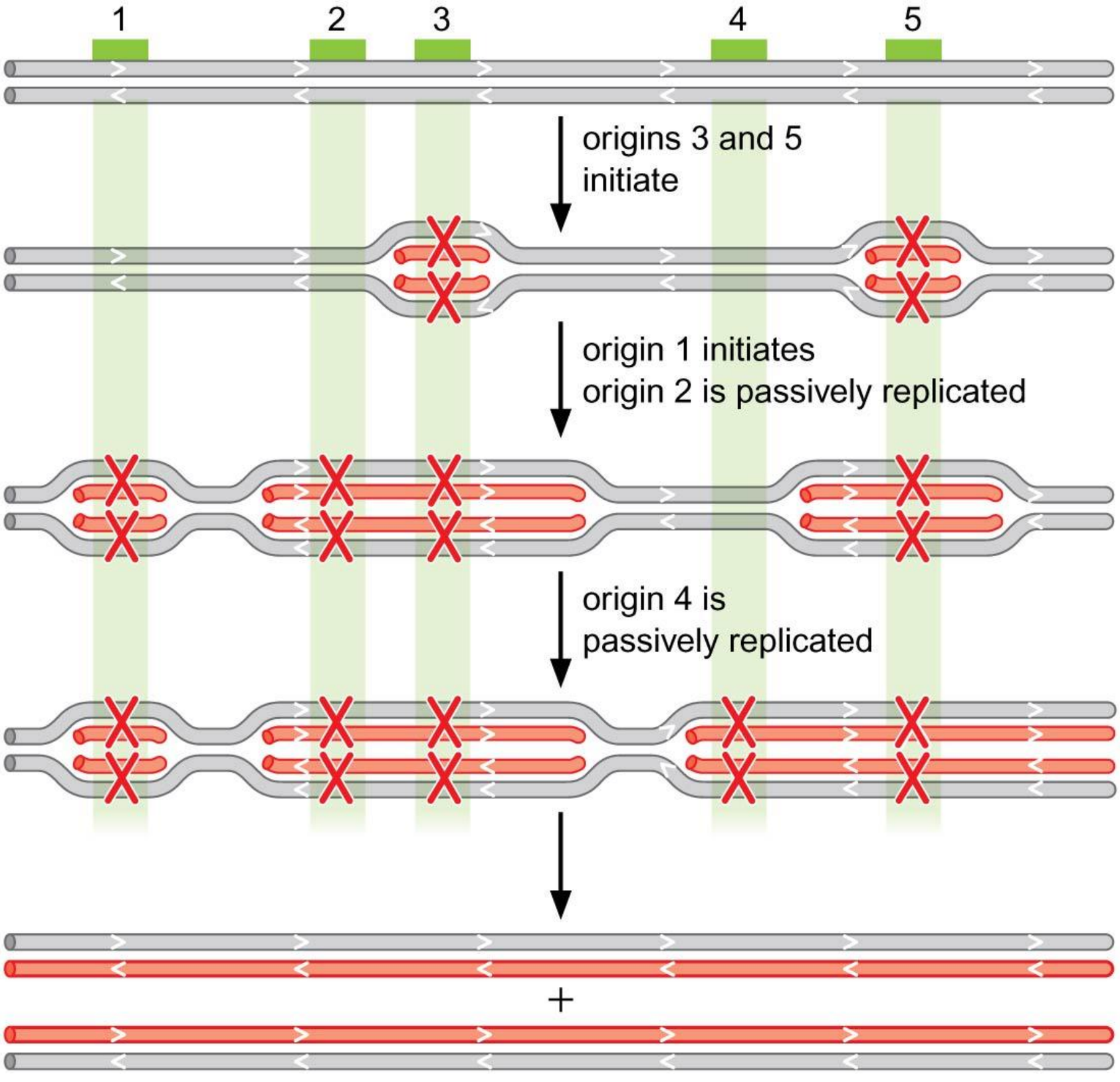


Figure 1-24 Molecular Biology of the Cell 6e (© Garland Science 2015)

- Origin of replication is called OriC in prokaryotes
- Only one origin of replication in prokaryotes.
- DNA replicated from one origin of replication is called a replicon.
- Only one replicon in prokaryotes.
- Termination of replication regions known as Ter regions

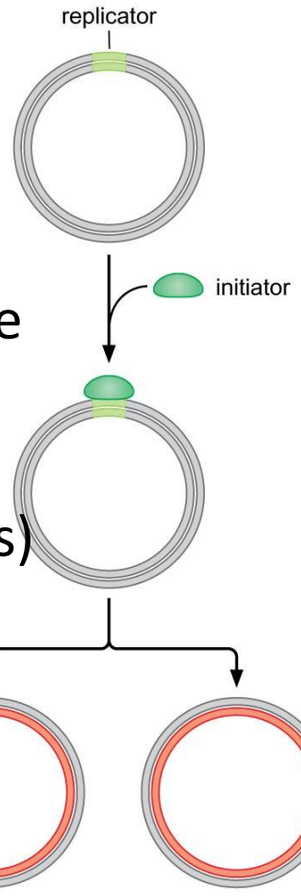
Where to begin replication? Eukaryotes



- Multiple replicons in eukaryotes
- Origins 3 and 5 fire first, followed by 1.
- Origins 2 and 4 are replicated by the elongation of replication forks from origins 3 and 5.
- Origins 2 and 4 are passively replicated.

Replicon Model of Replication Initiation

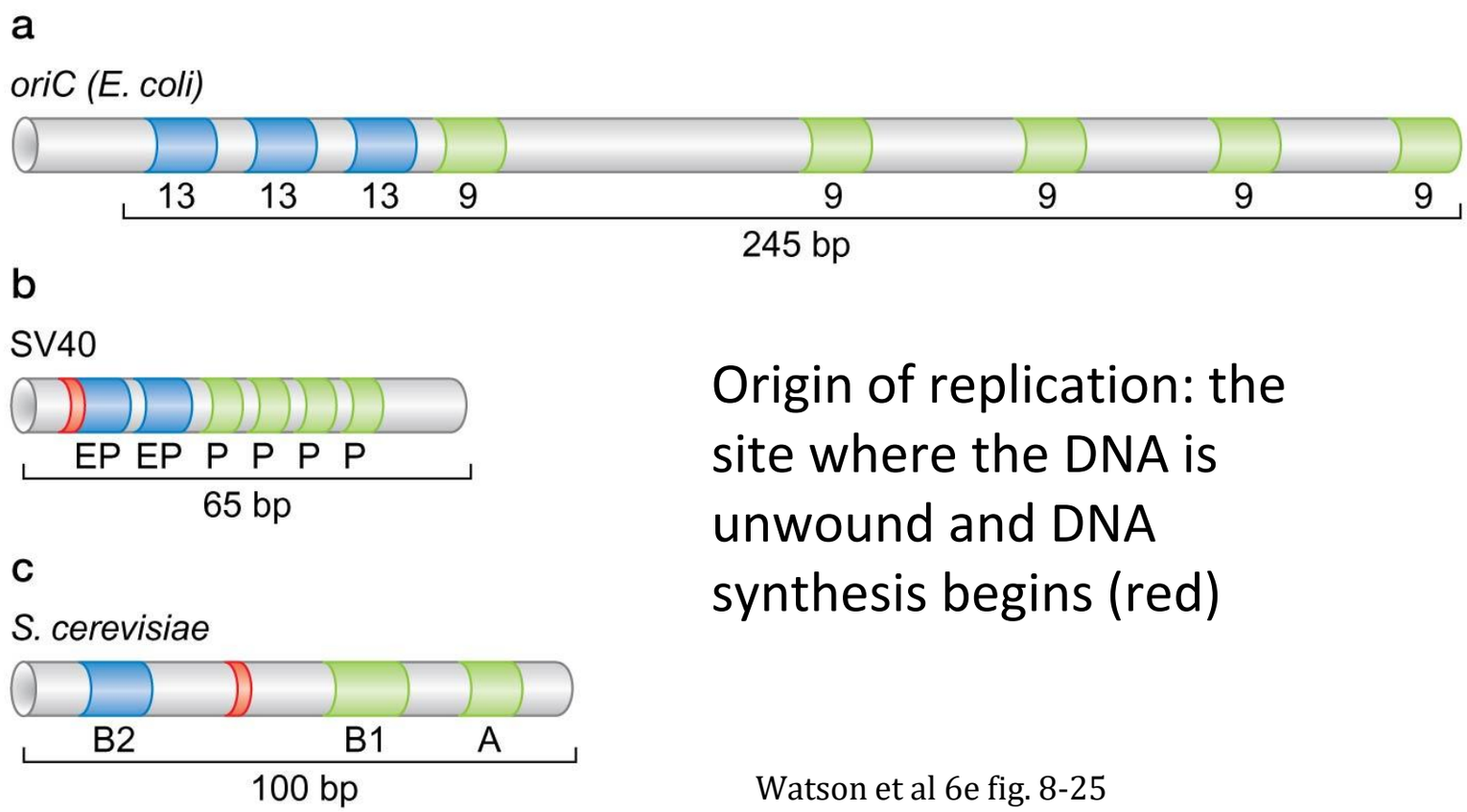
Figure 9-25



Watson et al 6e fig. 8-24

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Replicator: DNA sequence that is sufficient to direct the initiation of DNA replication. (blue, green and red regions)

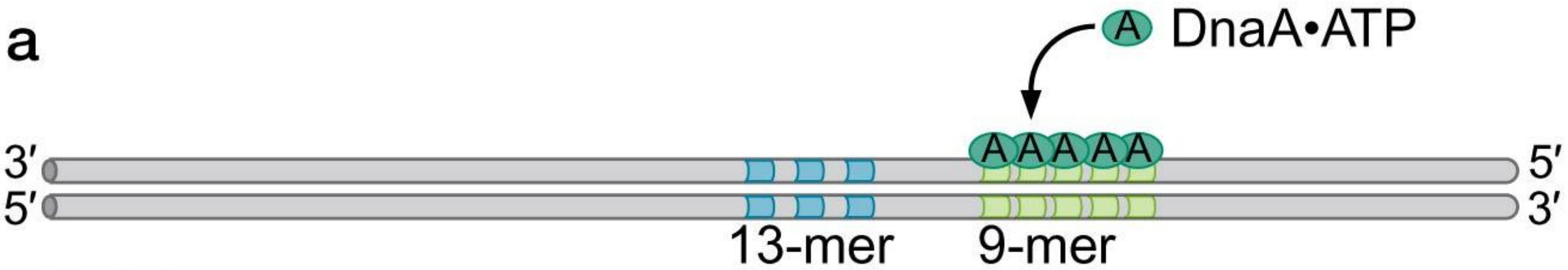


Watson et al 6e fig. 8-25

Origin of replication: the site where the DNA is unwound and DNA synthesis begins (red)

- Green, red and blue regions together represent the complete replicator region
- Green represents the binding sites for the initiator protein
- Blue represents the elements that facilitate DNA unwinding
- Red represents the site where DNA synthesis starts, i.e the origin of replication. These are defined sequences (This is upstream in E. coli and not shown in this figure)

Roles of DnaA, DnaB and DnaC (E.Coli)



- DnaA binds to double stranded 9mer regions.
- When DnaA is bound to ATP it can interact with DNA in 13mer region that results in it separating to single stranded DNA.
- Fig. 8.27 Watson 4e, 28.26 and 28.27 Stryer 8e

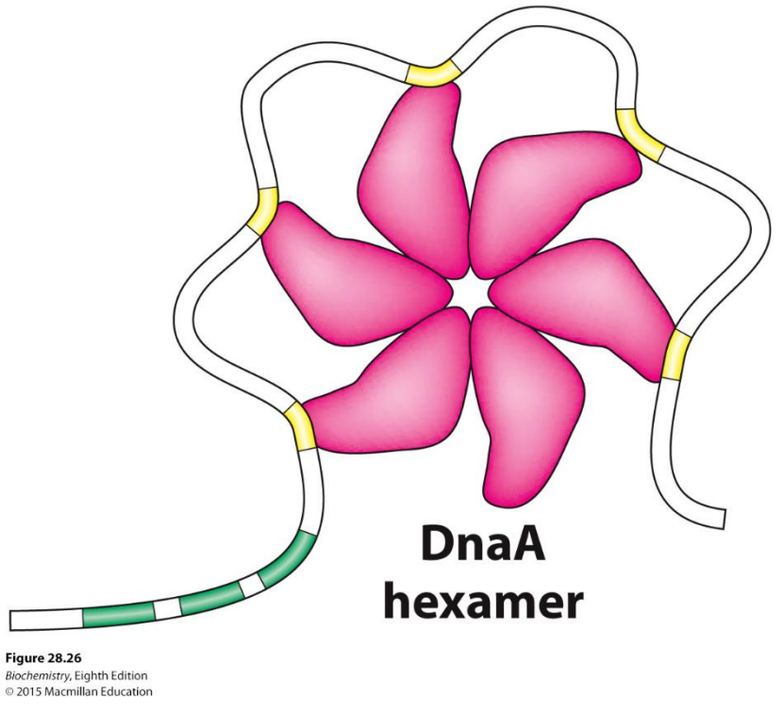
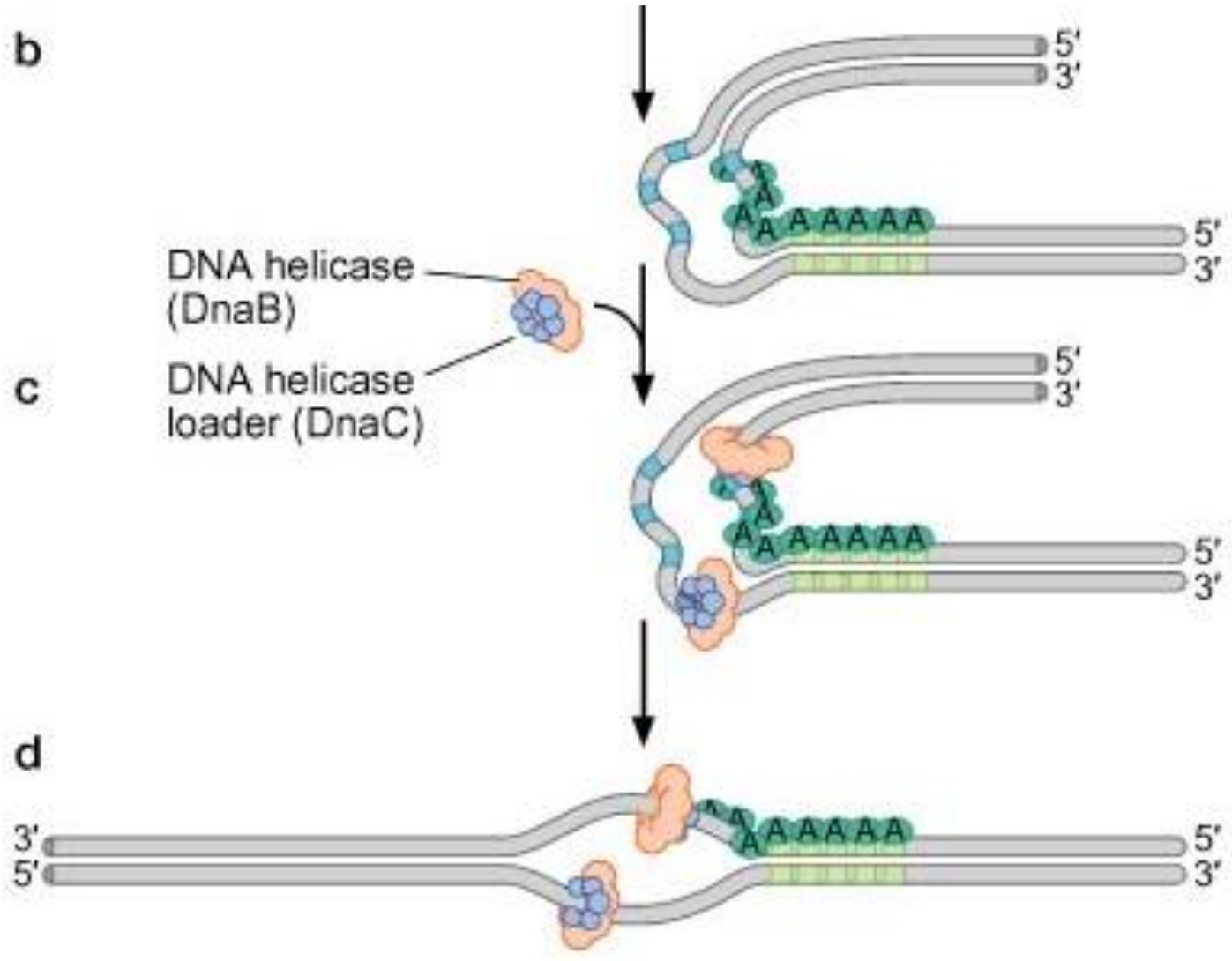


Figure 28.26
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Roles of DnaA, DnaB and DnaC



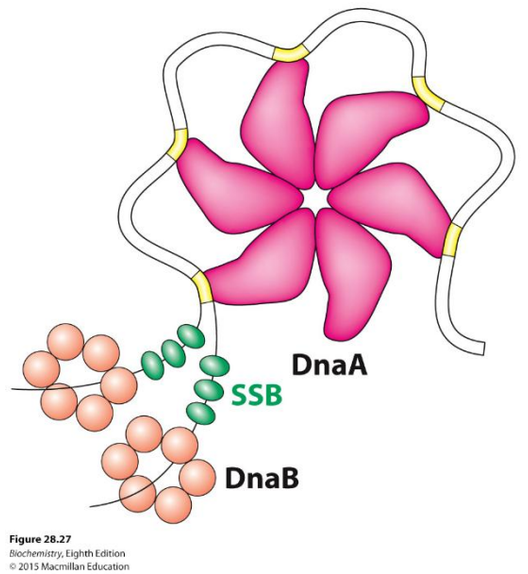
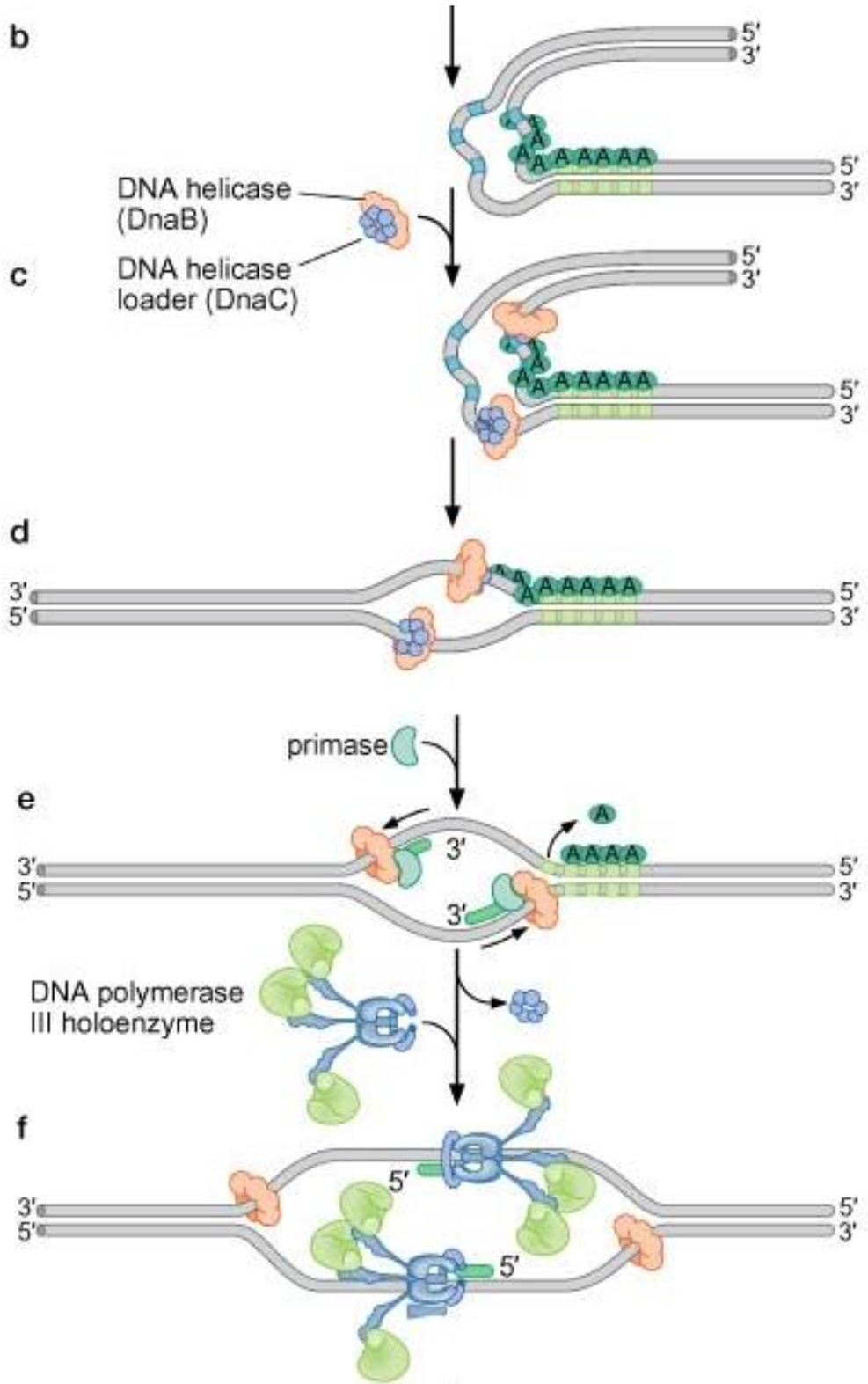


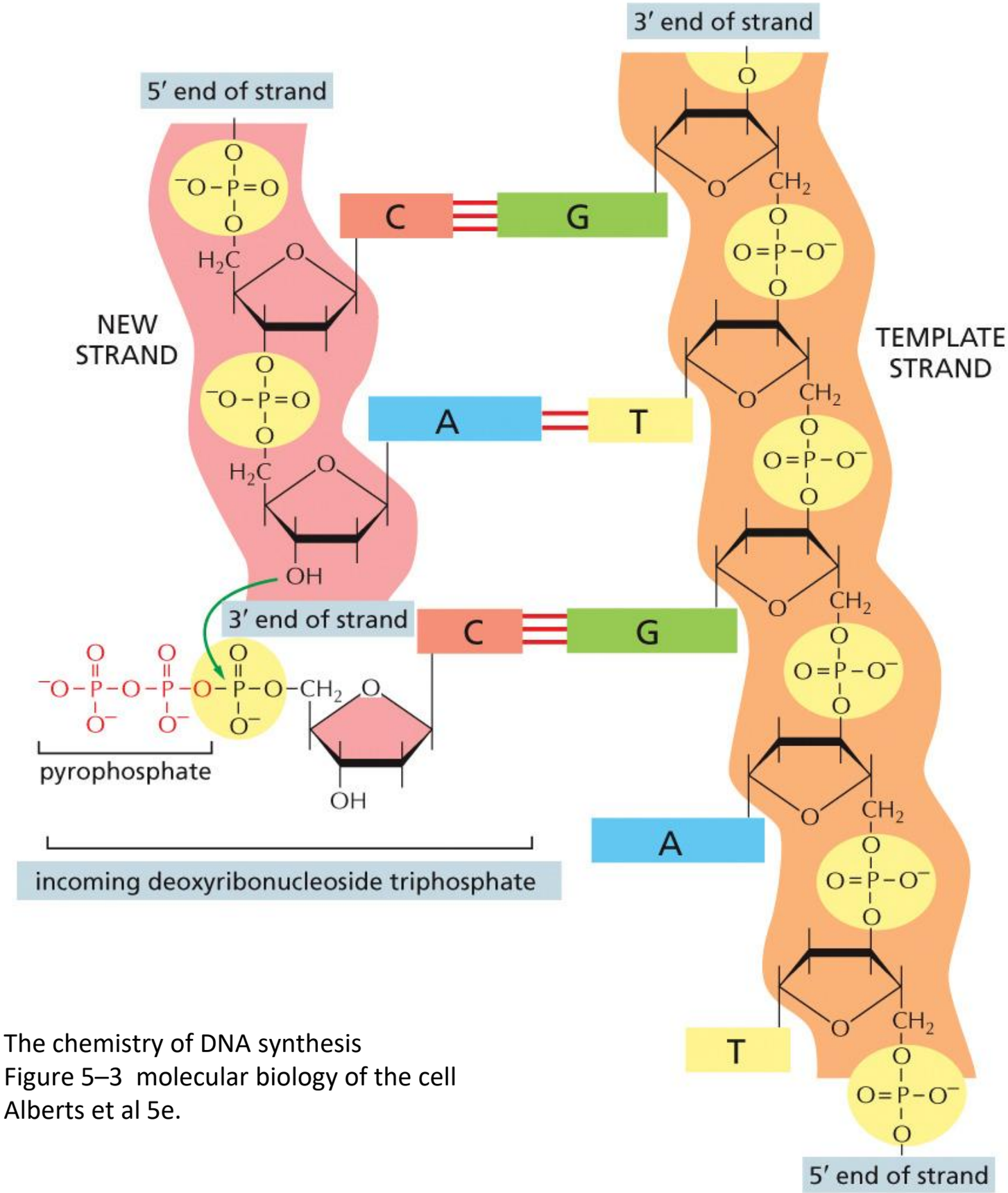
Figure 28.27
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- DnaA recruits complex of DnaB and DnaC, helicase and helicase loader respectively.
- Helicase is inactive when bound by the helicase loader and only activated when the helicase loader departs.
- Helicase in turn recruits primase, and primer synthesis causes the helicase to disassociate.

DNA replication



DNA synthesis proceeds in 5’ to 3’ direction



The chemistry of DNA synthesis
Figure 5–3 molecular biology of the cell
Alberts et al 5e.

Many anti cancer drugs target DNA synthesis; cancer cells are particularly susceptible since they are rapidly dividing.

Simultaneous synthesis on both strands

- DNA is synthesised 5’ to 3’ (but this is problematic as one DNA strand is anti-parallel)
- Continuous synthesis from a single RNA primer is only possible on the leading strand
- The lagging strand is synthesised as multiple **Okazaki fragments** that are joined together
- DNA polymerase 1 synthesises DNA to complete the lagging strand (i.e. connect the Okazaki fragments).
- Each Okazaki fragment has an RNA primer that is removed and replaced by DNA polymerase 1

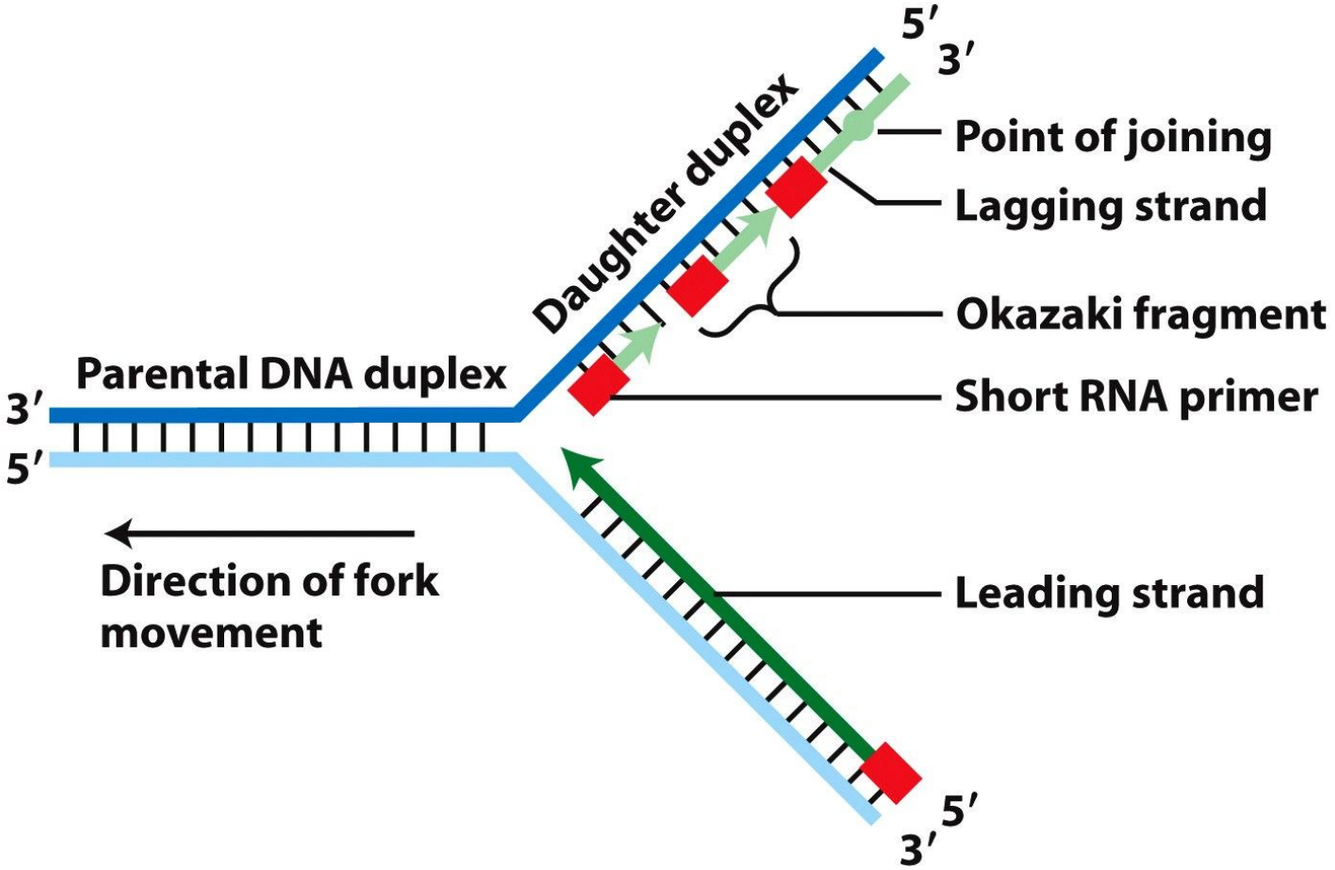


Figure 4-30
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DNA Replication proceeds until the entire genome is replicated

Prokaryotes

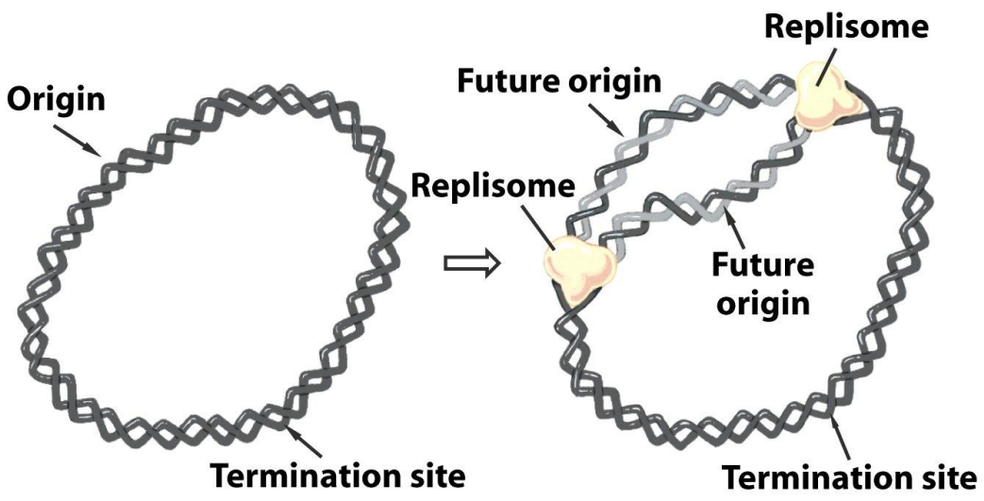
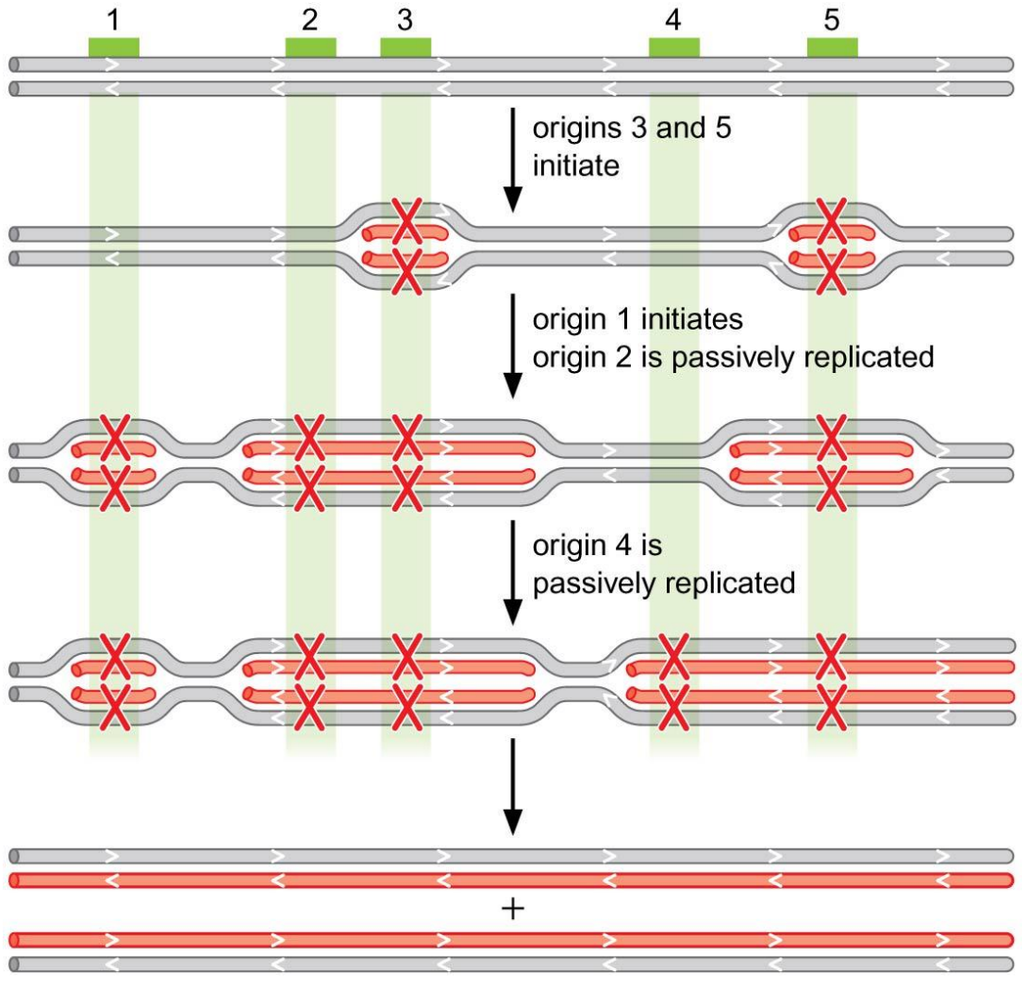


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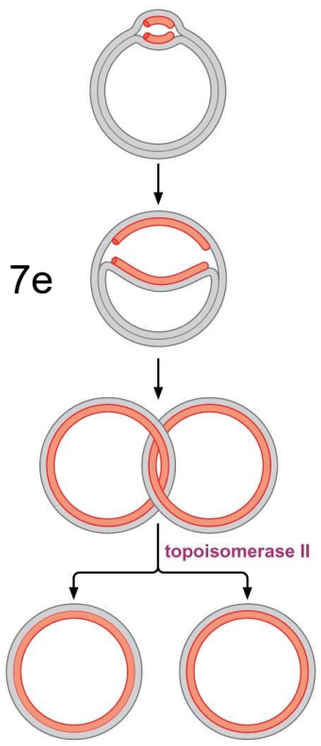
Eukaryotes

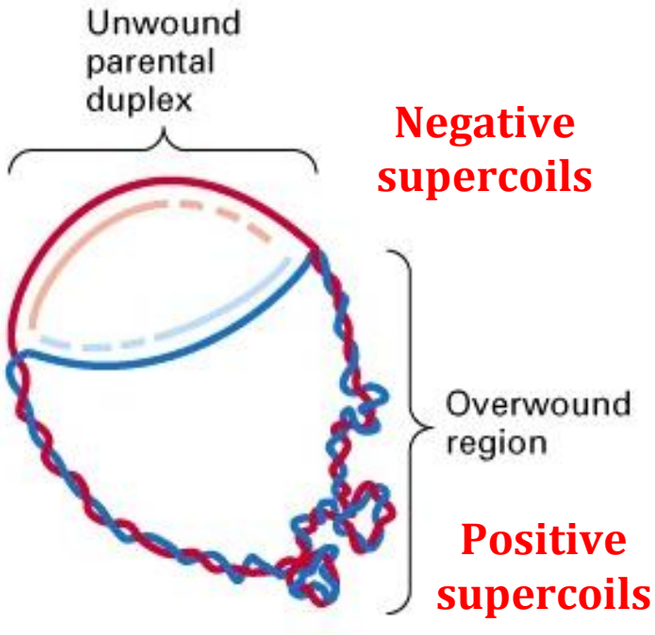


Termination in circular and linear chromosomes

- Catenanes-2 circular linked pieces of DNA

Fig 9.35 Watson et al 7e





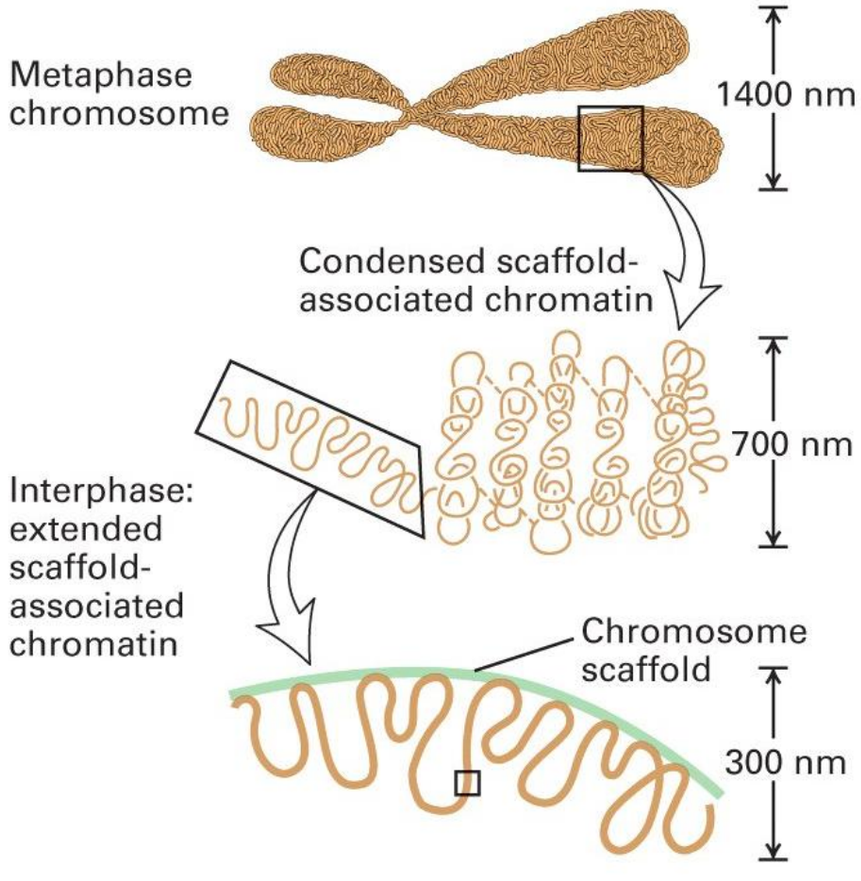
Negative supercoils

Positive supercoils

Topoisomerases:

Type I: relax DNA by nicking and closing one strand of duplex DNA

Type II: change DNA topology by breaking and rejoining double-stranded DNA



DNA polymerase: proofreading

- DNA replication is very accurate (10^{-9} to 10^{-10} errors per base pair per round of replication)
- DNA polymerase can proofread

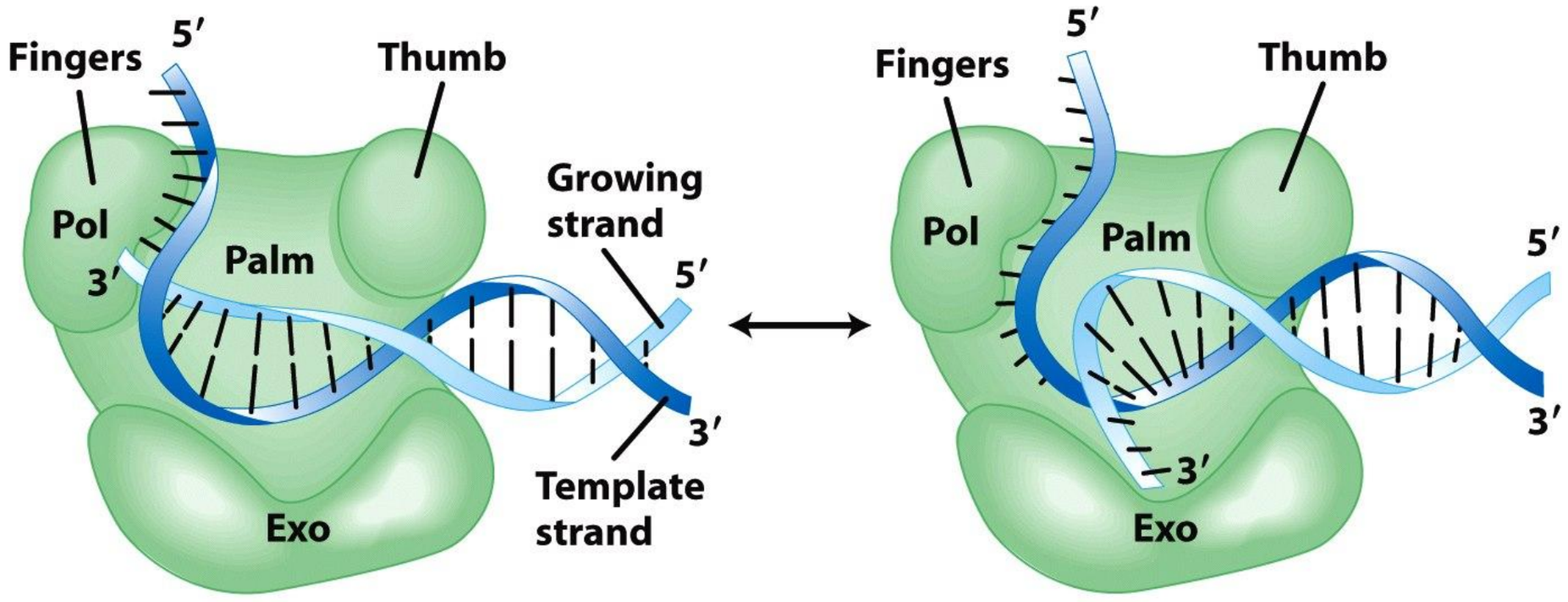


Figure 4-34
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- The physical shape of DNA polymerase resembles a hand.
- Incorrect base incorporation is sensed by physical distortion
- The DNA strand is moved to the Exonuclease domain, where the incorrect base is removed (3’ to 5’ exonuclease activity).
- The DNA strand moves back to the Polymerase domain.

Polymerase Chain Reaction (PCR) exploits DNA Replication

- 3 steps per cycle:
- Denaturation
- Annealing
- Extension
- Cycles repeat

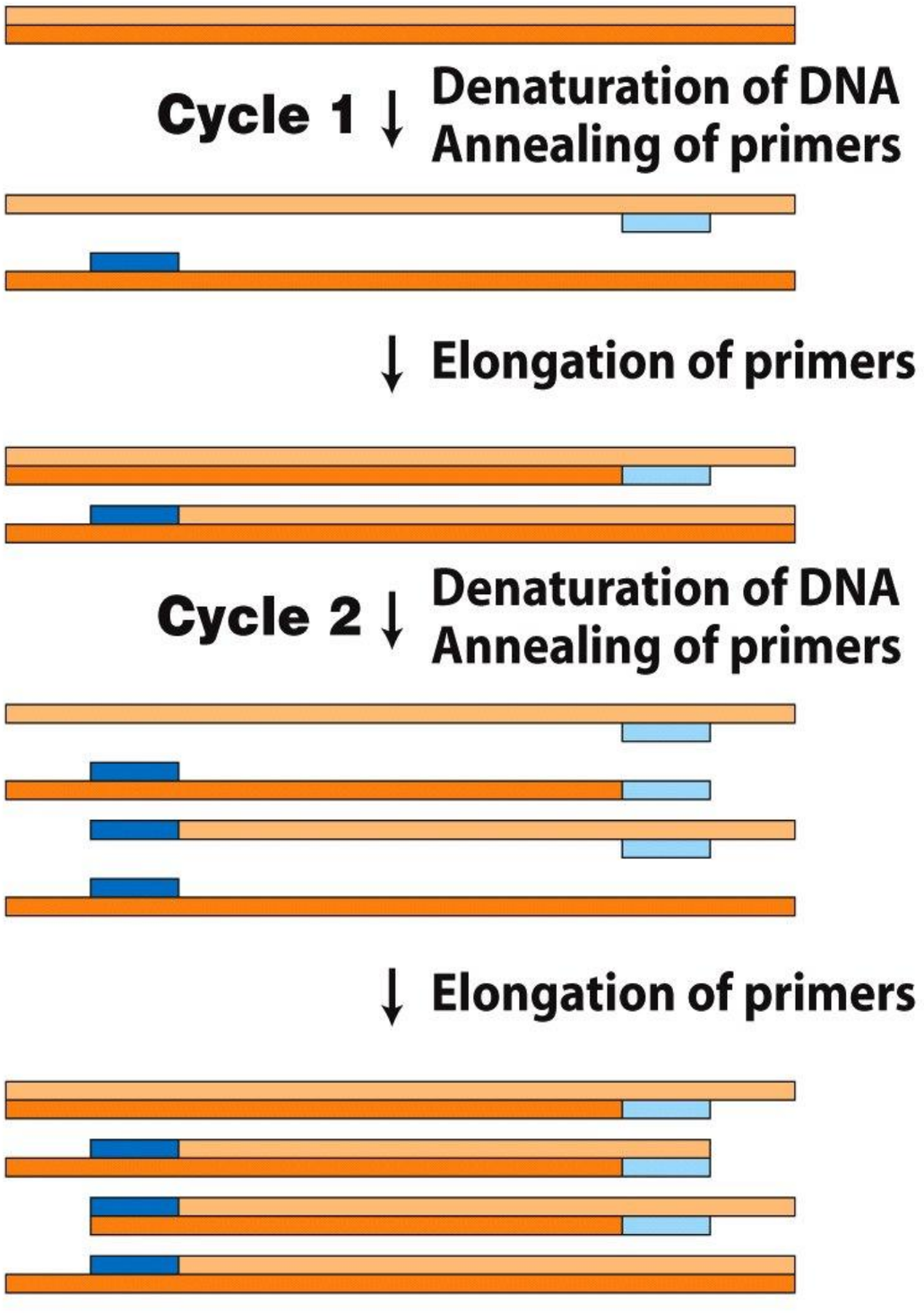


Figure 5-23 part 1
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Take home message

- DNA polymerases cannot melt or unwind duplex DNA

- DNA polymerases cannot initiate chains, but can only extend a pre-existing DNA or RNA strand

- DNA replication can only start at an origin of replication

- All DNA strands grow in a 5’ to 3’ direction

Question for next week

Replication: How fast is *E. coli* DNA made?

It takes ~42 minutes to replicate the single circular chromosome (4,639,221 bp) from the sole origin of replication.

What is the rate of fork movement?

Answer at the start of the next lecture!